

PROJECT DESIGN PHASE-II

Data Flow Diagram & User Stories

Date	3 October 2026	Team ID / Name	SWTID-2026-6919
Project Name	Script-Controlled ACL - Restrict Record Access Based on Field Value	Maximum Marks	4 Marks
Target Platform	ServiceNow Platform (Utah / Washington DC)	Document Version	v1.0 (Final)

1. DATA FLOW DIAGRAMS (DFD)

A Data Flow Diagram (DFD) visually models how information moves through a system, depicting entry points, processes, persistent data stores, and outputs. For the **Script-Controlled ACL** project, the DFD illustrates how user record requests interact with ServiceNow security mechanisms—specifically combining server-side *Access Control Lists (ACLs)* with *Before Query Business Rules* to enforce field-level security constraints seamlessly.

1.1 System Architecture & Data Flow Logic

1

User Record Request: Support Fulfillers or Department Users attempt to access record lists, forms, or global search queries.

2

Before Query Interception: A 'Before Query' Business Rule executes at database level, injecting filter conditions based on session user roles (`gs.getUserID()`) and field privacy flags.

3

Scripted ACL Evaluation: The Server-Side ACL Engine executes advanced JavaScript rules (`sys_security_acl`) on target fields like `current.u_confidential`.

4

Database Data Retrieval: The target record table (`u_incident_confidential`) returns authorized rows and field values without revealing restricted records.

5

Clean UI Rendering: The user interface displays filtered records cleanly without presenting 'Security constraints prevent viewing' warning banners.

1.2 DFD Component & Data Dictionary

Element Type	Component Name	Description / Function	Data Inputs & Outputs
External Entity	Fulfiller / User	Initiates record read/write requests via UI, reports, or global search.	In: Search query, Record ID Out: Filtered list, Form view
Process 1.0	Before Query BR Engine	Applies database-level filters before records are fetched from data store.	In: User session roles Out: SQL query filter conditions
Process 2.0	Scripted ACL Evaluator	Executes JavaScript rules (<code>answer = true/false</code>) on target fields.	In: <code>u_confidential</code> , Session ID Out: Boolean access decision
Data Store D1	ACL Store (<code>sys_security_acl</code>)	Stores record and field-level permission rules and advanced scripts.	In: Admin configurations Out: ACL rules & scripts

Element Type	Component Name	Description / Function	Data Inputs & Outputs
Data Store D2	Target Table (u_incident_confidential)	Stores transactional tickets with field values (State, Department, Confidential).	In: Authorized updates Out: Raw record datasets

2. USER STORIES & ACCEPTANCE CRITERIA

The table below details the complete set of Agile User Stories, functional requirements, acceptance criteria, priority levels, and assigned release sprints for the project.

User Type	Epic / Module	ID	User Story / Task	Acceptance Criteria	Priority	Release
ServiceNow System Administrator	ACL Configuration & Scripting	USN-1	As an Admin, I can configure a record and field-level read ACL with 'Advanced' scripting enabled, so that access is evaluated dynamically based on field values.	ACL script evaluates <code>current.u_confidential</code> against session user roles (<code>gs.getUser()</code>), setting <code>answer= true/false</code> accurately.	High	Sprint-1
ServiceNow System Administrator	ACL Configuration & Scripting	USN-2	As an Admin, I can configure a matching 'Before Query' Business Rule on the target table, so that unauthorized records are filtered out at database execution time.	List views suppress restricted rows silently without displaying 'Security constraints prevent viewing' warning banners.	High	Sprint-1
ServiceNow System Administrator	Validation & Testing	USN-3	As an Admin, I can use 'Impersonate User' to test access permissions across different departmental roles, so that security policies are validated prior to production.	Impersonated non-authorized users cannot view or search restricted records across forms, lists, or global search.	High	Sprint-1
Support Fulfiller / Agent	Record View & Operation	USN-4	As a Fulfiller, I can view and edit records assigned to my department, while confidential records belonging to	Form fields for confidential tickets are rendered read-only or hidden based on active user departmental assignment.	High	Sprint-1

User Type	Epic / Module	ID	User Story / Task	Acceptance Criteria	Priority	Release
			other teams are hidden.			
Support Fulfiller / Agent	Search & Reporting Experience	USN-5	As a Fulfiller, I can execute global search queries without seeing restricted record summaries in search indexes.	Global search results and report widgets display only records where read access ACL evaluates to true.	Medium	Sprint-2
Department Manager	Confidentiality Management	USN-6	As a Department Manager, I can toggle the 'Confidential' field on sensitive tickets so that access is immediately restricted to authorized role holders.	Updating <code>u_confidential = true</code> dynamically updates row access permissions upon saving the record.	High	Sprint-1
Department Manager	Dashboard Privacy	USN-7	As a Department Manager, I can share operational reports without exposing confidential ticket metrics to unauthorized teams.	Reporting widgets apply table security constraints automatically, maintaining strict data privacy.	Medium	Sprint-2
Compliance & Security Auditor	Security Auditing	USN-8	As an Auditor, I can review security access logs and field ACL definitions, so that compliance with enterprise privacy standards is verified.	All access evaluations are fully traceable, and unauthorized read/write attempts are blocked and logged.	High	Sprint-2